The contact-edge selector: why $\theta_c = \inf \mathcal{F}$ fails,

and the closed-form tangency residual that replaces it

SpectralKM derivations

1 The symptom

Measured against the published data of [1], this model overpredicts the contact time by roughly a factor of two: t_c/t_σ of 4.9–6.1 against an experimental 2.4–4.0, and nearly *flat* in We where the experiment decreases. Three candidate causes were tested and eliminated:

Candidate	Test	Outcome
metric definition	`threshold_contact_time` vs. `contact_time`	real but small: 0.6 of ~ 2.0
drop viscous closure	`:reid` vs. `:lamb`	differ in the 4th digit
bath radius (reflected waves)	`debug_bath_radius_sensitivity.jl`	flat: $4.9402, 4.9613, 4.9614, 4.9614$
	$b = 6, 12, 20, 30,\ M \propto b$	for $b = 6, 12, 20, 30$

The bath-radius result deserves a note: $b = 6$ is flagged as unvalidated in the open-questions section of `provenance.tex` and in `paper-formulation.tex`, with a prior measurement of 11% CoR movement between $b = 6$ and $b = 4$. It is a genuine open question and it is *not* the cause of this symptom.

2 What is actually wrong

Tracing a contact (`debug_contact_exit_trace.jl`, `debug_feasibility_collapse.jl`):

- the net force $f \leq 0$ on **0 of 221** steps – it decays asymptotically to 1.4×10^{-8} from above and never reverses, so the $f \leq 0$ exit test never fires and contact terminates only at the θ_c floor;
- 63% of the contact duration occurs *after* the droplet has begun rising;
- at $t = 0.009$ the non-intersection predicate is false for all $\theta < 0.0492$ and $\inf \mathcal{F} = 0.0492$; one step later, at $t = 0.010$, it is true even at $\theta = 10^{-4}$, $\inf \mathcal{F}$ collapses by a factor of 200, and the patch never recovers.

The search is not at fault. A brute-force scan (`debug_selector_rate_limit.jl`) shows `select_theta_c` returning within 0.3–4% of the true $\inf \mathcal{F}$ at every step. The *rule* is at fault.

3 The degeneracy, stated precisely

Write the gap as $C(\theta,\tau) = \eta(r(\theta,\tau),\tau) - z_{cm} + z_d(\theta,\tau)$. The kinematic match imposes $C = 0$ on $[0,\theta_c]$; physical consistency requires $C < 0$ beyond the edge. Continuing just outside, $C(\theta) \simeq a\,(\theta - \theta_c)$ with

$$a(\theta_c) := \partial_\theta C\big|_{\theta_c}, \tag{1}$$

so non-overlap is the inequality $a(\theta_c) \le 0$ and the feasible set is $\mathcal{F} = \{\theta_c : a(\theta_c) \le 0\}$. The two rules are then

$$\text{current: } \theta_c = \inf \mathcal{F}, \qquad \text{tangency (3): } a(\theta_c) = 0. \tag{2}$$

> If $a(\theta_c) < 0$ for every $\theta_c > 0$ – the measured situation from $t = 0.010$ onward – then $\mathcal{F} = (0, \theta_{\max}]$ and $\inf \mathcal{F} = 0$, *regardless of the shape or magnitude of* a.

This is the whole failure. The infimum rule uses only the **sign** of a and never its magnitude, so it returns the same answer whether the surfaces are barely separated or widely so. It carries information only while the constraint is *binding*; the moment non-intersection goes inactive, it degenerates to $\theta_c = 0$ – not because the physics says the patch vanishes, but because the rule has nothing left to determine it with. Verified symbolically in `derivations/cas_tangency_residual.py`, which also exhibits an $a < 0$ example where $\inf \mathcal{F} = 0$ while $\arg\min |a|$ is interior.

Two further facts from the scan bear on the rule's stated justification. Positivity ($p > 0$) holds *at* the collapsed patch, so complementarity – which `select_theta_c`'s docstring cites as its basis – does not forbid it. And the band's upper edge is set by $f < 0$ and $p < 0$, i.e. by suction, not by `eq:check-monotone-r`. The rule's original evidence was a single measurement at one step at onset, precisely where the constraint *is* active and the rule does work.

4 The closed-form tangency residual

`eq:tangency-selector` is $T = \partial_\theta C|_{\theta_c}$, currently evaluated in `src/residual.jl` by a centred finite difference with $h = 10^{-6}$.

A claim withdrawn. An earlier draft of this note asserted that differencing a Fourier–Bessel sum at $h = 10^{-6}$ "discards roughly half the available digits" and was therefore unfit to root-find. *That was wrong, and the test written to check it said so*: against a Richardson-extrapolated reference at $h = 10^{-3}$, the existing finite-difference residual agrees to `rtol` $= 10^{-8}$ at $\theta = 0.25, 0.6, 1.1$ (`test/test_selector.jl`). Accuracy is not the problem. The closed form below is still worth having — it removes a step-size parameter, and it differentiates cleanly under `ForwardDiff` where a difference of a difference would not — but the case for it is structural, not numerical, and the fix to the selector does not depend on it. With $\xi(\theta) = 1 + \sum_l \beta_l P_l(\cos\theta)$, $r = \xi \sin\theta$, $z_d = -\xi\cos\theta$ and $\eta(r) = \sum_m a_m J_0(k_m r)$, the chain rule gives

$$\boxed{T = \eta'(r)\left[\xi' \sin\theta + \xi\cos\theta\right] + \left[\xi\sin\theta - \xi'\cos\theta\right]} \tag{3}$$

with

$$\xi'(\theta) = -\sin\theta \sum_l \beta_l P_l'(\cos\theta), \qquad \eta'(r) = -\sum_m a_m k_m J_1(k_m r). \tag{4}$$

(3) is verified against symbolic differentiation of the composed expression (difference exactly zero), and against the existing numerical residual in `test/test_selector.jl`.

5 The pole degeneracy is physical, not algebraic

$T \equiv 0$ at $\theta = 0$ and $\theta = \pi$ for any state – `eq:tangency-degenerate`. Worth recording *how* the CAS established this, because the first attempt was wrong: with η left as an abstract function the CAS correctly **refused**, returning $T(0) = \xi(0)\,\eta'(0)$, since nothing in the chain rule makes $\eta'(0)$ vanish. It vanishes because of **axisymmetry**, through $J_1(0) = 0$; and ξ' carries a factor $\sin\theta$, so it vanishes at both poles too. Substituting the concrete expressions, $T \to 0$ at both poles for $l = 2$ and $l = 3$ and arbitrary $\beta_l, a_m, k_m, z_{cm}$.

Two implementation consequences follow. A seed is required at onset, which `onset_theta_c` already supplies from the geometric crossing $C = 0$; and the residual must be root-found *away* from $\theta = 0$, never bracketed from it.

6 The replacement rule

Select θ_c as a root of (3), retaining `eq:check-nonintersect` and `eq:check-monotone-r` as a *feasibility filter* over candidates – which is what `paper-formulation.tex` describes, and what [1] and [2] do. Because $\eta(r(\theta))$ is a sum of oscillatory J_0 terms, T can change sign more than once on $(0, \pi)$, so a plain bisection on a sign change is not safe; the argmin form of `eq:theta-c-argmin` over admissible candidates is used instead, seeded from the previous step and, at onset, from `onset_theta_c`.

7 Scope

This note establishes that the current rule is degenerate and derives the residual that replaces it. It does *not* establish that tangency selection reproduces the experimental t_c: that is a measurement, made after implementation, against `data/experiments/contact_time_vs_we.csv` and the $r_c(t)$ series of Figure 6(b). Nor does it revisit the upper edge of the admissible band, where $f < 0$ and $p < 0$ already exclude wide patches; whether that exclusion is itself too aggressive is a separate question.

References

[1] L. F. L. Alventosa, R. Cimpeanu, D. M. Harris, *Inertio-capillary rebound of a droplet impacting a fluid bath*, JFM (2023).

[2] C. A. Galeano-Ríos, P. A. Milewski, J.-M. Vanden-Broeck, *Non-wetting impact of a sphere onto a bath*, JFM (2017).